

Supporting Information for “Transferable Class-Correction Refinement from Multiple Rotational Data Sets”

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Reproducible Anhydride Runs

The comparison tables and plotting fragments in the main text were generated from the repository command

```
python -m merlino_core.cli semiexp-ensemble-paper
--job <anhydrides ensemble job>
--paper-dir <JPCL paper directory>
--outdir <analysis directory>
```

The command writes separate reports for the no-prior, soft-prior, and hard-constraint variants, together with CSV, JSON, and L^AT_EX summaries.¹ It also writes the prior-scan and leave-one-molecule-out CSV files used to generate the corresponding tables in the main text.

Reproducible Glycine Runs

The glycine conformer comparison was generated from the existing legacy MSR benchmark inputs. The element/value-window run used

```
python -m merlino_core.cli semiexp-ensemble
--job <glycine conformer ensemble job>
--outdir <glycine ensemble output>
```

and the continuous synthon-typed run used the corresponding job with Z_{eff} prototypes and a threshold of 0.035. The threshold scan was generated with

```
python -m merlino_core.cli semiexp-ensemble-synthon-scan
--job <glycine synthon ensemble job>
--outdir <threshold scan output>
--threshold 0.010 --threshold 0.035 --threshold 0.075
--threshold 0.100 --threshold 0.150 --threshold 0.250
```

Coordinate-Class Policy

Stretches are matched by unordered atom pair and can be further separated by geometric ranges, for example carbonyl and single C–O distances. Bends preserve the central atom and canonicalize only the terminal atoms. Torsions are matched by the unordered pair of central atoms. Out-of-plane coordinates are matched by the central atom in $U(\text{center}, a, b, c)$. Synthon-based typing uses the continuous effective atomic number Z_{eff} as a local-environment descriptor; two atoms are assigned to the same type when their Z_{eff} values lie within the declared threshold.

Current Test Cases

The present working set contains maleic, phthalic, and succinic anhydrides, plus glycine conformers I and II. The anhydrides test a homologous class with different isotopic coverage and different needs for regularization. The glycine pair tests continuous synthon atom typing across conformers with different geometries and atom orderings. Additional large-molecule demonstrations should be added only after these two comparisons have been finalized.

References

- (1) Barone, V. MORPHEUS multi-molecule class-correction refinement working implementation. 2026; Working repository benchmark and software manuscript scaffold.